Extension of the RSA Production Model

Mechanism-level exploration and the 2×2 tradeoff

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Where we start: the paper-reported model

Incremental recursive speaker, hierarchical per-participant α offsets. Theory-driven: minimal parameters with clear motivation.

The 2×2 story (memo §1):

Model	ELPD LOO	Δ ELPD	dSE	Δ/dSE	*p*
Inc. recursive	-7180	0	—	—	—
Inc. static	-7199	-19	1.84	-10.28	$< .001$
Global static	-7626	-446	40.27	-11.08	$< .001$
Global recursive	-7628	-448	40.69	-11.02	$< .001$

- **Speaker architecture dominates**: incremental $\gg$ global ($\Delta/\text{dSE} \approx 11$).
- **Semantic regime within incremental**: recursive $>$ static ($\Delta/\text{dSE} = 10.28$).
- **Semantic regime within global**: null (global recursive slightly *worse*).

But: the fit is poor

Production data = categorical utterance choice among 15 types, in 9 conditions × 2 sharpness levels.

- $R^2 = 0.346$ on the 90 (condition × utterance) cells.
- L1 aggregate residual $= 3.17$ across cells with empirical $\geq 2\%$.
- Lapse $\varepsilon = 0.22$ — nearly a quarter of responses treated as uniform noise.

Seven systematic misfits (memo §5.3):

Condition	Sharpness	Utterance	Emp	Model	Δ
colour-sufficient	high	C	.716	.403	−.313
size-sufficient	low	SF	.365	.081	−.284
colour-sufficient	low	C	.628	.402	−.225
both-necessary	low	C	.009	.222	+.214
size-sufficient	high	SFC	.022	.211	+.189
both-necessary	high	C	.000	.175	+.175
size-sufficient	low	SFC	.032	.187	+.155

Diagnosis (memo §5.3)

Residuals decompose along two independent axes.

1. Context-dependence of 1-word C:

- Under-predicted by $\sim$ .25–.31 in colour-sufficient contexts
- Over-predicted by $\sim$ .18–.21 in both-necessary contexts
- No single α_C can fix both.

2. Length residuals show saturation, not level shift:

- 2-word (SF) under-predicted by $\sim$ .28 in blurred size-sufficient
- 3-word (SFC) over-predicted by $\sim$ .17 in size-sufficient
- No monotone change to a linear γ fixes both.

These diagnostics motivate the mechanism track.

Extension timeline (dc subset, $N = 3196$)

Conditions `erdc, zrdc, brdc` — size $\times$ colour distinguishing, form redundant. Three `relevant_property` values inside: `first` (size-alone-sufficient), `both` (both needed), `second` (colour-alone-sufficient).

Step	Mechanism	Parameters	Rationale
ext-v1	per-dim α, γ, ε	$\alpha_D, \alpha_C, \alpha_F, \gamma, \varepsilon$	dim-salience asymmetry
v5a	condition-gated colour boost	λ_C	Pechmann 89; Rubio-Fernández 16
v5b	saturating length bias	γ_1, γ_2	overspec not monotone in length
v5 (F1)	+ csuff length offsets	$\delta\gamma_1, \delta\gamma_2$	bare-C when colour identifies
v5 (C1)	+ canonical-order penalty	μ_{nc}	Sproat 91; Cinque 94; Scott 02
v5 (F2)	+ sharpness-dependent length	η_1, η_2	overspec more common under blur

Flag correction: `COLOUR_SUFFICIENT_CONDITIONS` was first (*ercf*, *zrcf*, *brcf*). Corrected to (*ercf*, *zrdc*) — only conditions where colour alone identifies the target.

Attempt 1: ext-v1 (already in memo §4.1)

Per-dimension α + explicit γ + lapse ε.

	ELPD	vs baseline
baseline	-7157	0
ext-v1	-6387	$+770$

	R^2	L1
baseline	.35	3.17
ext-v1	.60	3.17

$\alpha_D = 6.94, \alpha_C = 1.84, \alpha_F = 1.88, \gamma = 2.32, \varepsilon = 0.22.$

But: seven memo §5.3 misfits persist; $\varepsilon = 0.22$ uncomfortably high.

Attempt 2a: λ_C alone (v5a)

Step-0 logit boost for **C** when `is_colour_sufficient` $= 1$.

	ELPD	Δ vs ext-v1	λ_C posterior
ext-v1	-6387	0	—
v5a	-6265	$+122$	**3.4** (95% CI 2.8–4.0)

- Strong positive λ_C: colour gets extra first-step pull in colour-sufficient trials.
- second/sharp C: emp .716, was .403, now .55.
- CF still too high in second conditions: boost spreads to C-initial compounds (`CF`, `CDF`). Not enough by itself.

Attempt 2b: saturating γ alone (v5b)

Replace linear $\gamma \cdot k$ with $\gamma_1 \, \mathbb{I}[k \geq 1] + \gamma_2 \, \mathbb{I}[k \geq 2]$.

	ELPD	Δ vs ext-v1
ext-v1	-6387	0
v5b	-6381	$+6$

Essentially no improvement without λ_C.

The two mechanisms are synergistic: γ-saturation only matters after C-initial mass is correctly apportioned.

Attempt 3: F1 — condition-dependent length (v5 with $\delta\gamma$)

Let the length bias *shrink* on colour-sufficient trials ($\gamma_{1,\text{eff}} = \gamma_1 + \delta\gamma_1 \cdot \texttt{is\_csuff}$).

	ELPD	Δ vs ext-v1	notes
ext-v1	-6387	0	
v5a	-6265	$+122$	λ_C only
v5 (F1)	-5933	$+454$	$\lambda_C + \delta\gamma$

Posteriors: $\lambda_C = 5.0$, $\gamma_1 = 2.56$, $\gamma_2 = 0.92$, $\delta\gamma_1 = -1.89$, $\delta\gamma_2 = -3.51$, $\varepsilon = 0.16$.

Effective γ on colour-sufficient trials: $0.67, -2.59$ — 3-word heavily penalised, short utterances favoured. $R^2 = .67$, L1 $= 2.70$.

Remaining puzzle: model still over-predicts `DC` and `DFC`, under-predicts `DCF` in both-necessary and first-conditions.

Attempt 4: C1 — canonical-ordering penalty

Add $\mu_{\mathrm{nc}} \cdot \mathbb{I}[\text{utterance violates size} < \text{colour} < \text{form}]$.
Non-canonical set: `CD`, `CDF`, `CFD`, `DFC`, `FD`, `FDC`, `FC`, `FCD`.

	ELPD	Δ vs ext-v1	notes
v5 (F1)	-5933	$+454$	$\lambda_C + \delta\gamma$
v5 (C1)	-5323	$+1063$	$+\mu_{\mathrm{nc}}$

$\mu_{\mathrm{nc}} = -5.07$ — speakers strongly prefer canonical English ordering.

ε drops to 0.10.

$R^2 = .91$, L1 $= 1.34$.

7 of 8 user-flagged cells fixed: `DCF`, `DF`, `DC`, `DFC`, `C` residuals largely resolved.
Side-effect: on `first/sharp`, bare `D` under-predicted (.082 vs emp .269) and `DF` over-predicted.

Attempt 5: F2 — sharpness-dependent length

Add η_1, η_2 offsets active on `is_sharp` trials.

	ELPD	Δ vs ext-v1	R^2
v5 (C1)	-5323	$+1063$	.91
v5 (F2)	-5294	$+1093$	.94

Posteriors: $\eta_1 = -0.61$, $\eta_2 = -0.34$ (both informatively negative).
Effective γ on sharp: $1.71, 1.24$ vs blurred: $2.32, 1.58$.

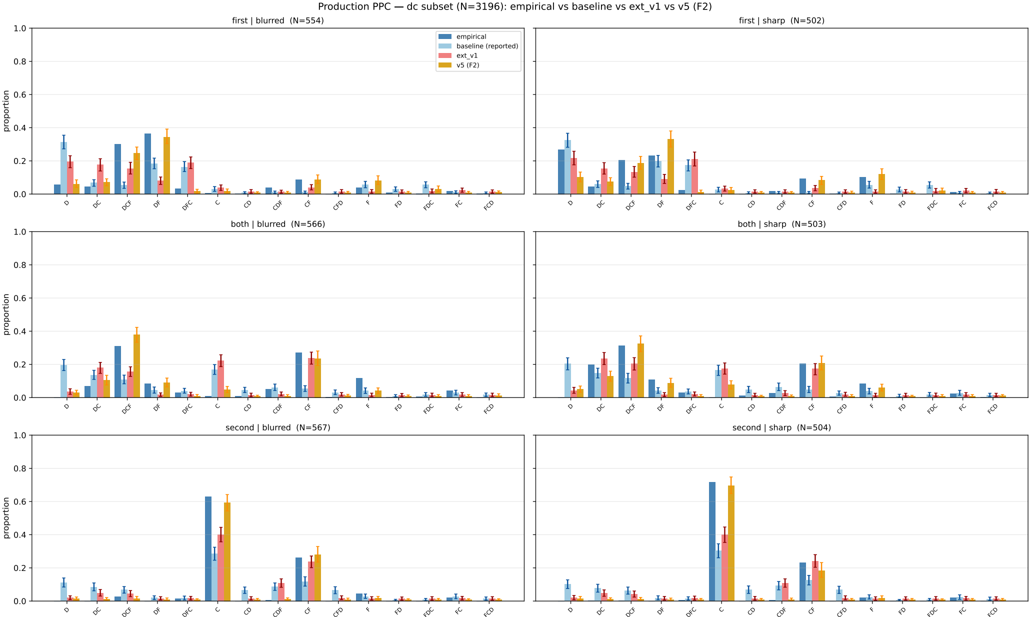
ε stays at 0.10; 0 divergences throughout; all $\hat{R} \leq 1.02$.

Stubborn residual: `first/sharp D` still under-predicted (.269 vs .102). Likely strategy heterogeneity (some participants say bare `D`, others hedge with `F`).

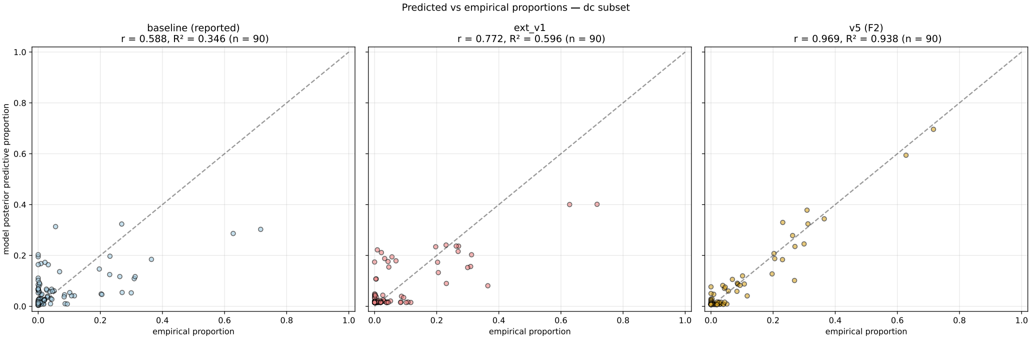
Final v5 (F2) parameters

Parameter	Prior	Posterior mean (95% CI)	$\hat{R}$
α_D	HalfN(5)	(hier. w/ δ_p)	≤ 1.01
α_C	HalfN(5)	(hier.)	≤ 1.01
α_F	HalfN(5)	(hier.)	≤ 1.01
λ_C	N(0,1)	3.09 (2.67, 3.55)	1.00
γ_1	N(0,1)	2.32 (2.16, 2.49)	1.00
γ_2	N(0,1)	1.58 (1.43, 1.74)	1.01
$\delta\gamma_1$	N(0,1)	-2.16 (-2.33, -1.97)	1.00
$\delta\gamma_2$	N(0,1)	-0.58 (-1.21, $+0.04$)	1.00
η_1	N(0,1)	-0.61 (-0.77, -0.43)	1.00
η_2	N(0,1)	-0.34 (-0.51, -0.15)	1.00
μ_{nc}	N(0,1)	-5.08 (-5.61, -4.56)	1.01
$\log\beta$	N(0,0.5)	-0.17	1.00
ε	Beta(1,50)	0.10 (0.09, 0.12)	1.00
τ	HalfN(0.2)	0.62 (0.51, 0.73)	1.00

PPC barplot — baseline vs ext-v1 vs v5 (F2)



Correlation — predicted vs empirical ($n = 90$)



R^2 climbs: baseline .35, ext-v1 .60, **v5 (F2)** .94.

The 2×2 question: does the paper story survive?

We re-ran the 2×2 (speaker architecture $\times$ semantics regime) with F2 mechanisms in all four cells, then again with freely-sampled parameters in the global cells.

Three parameterisations to compare:

1. **Paper-reported (memo §1)**: minimal, theory-driven.
2. **F2-fixed globals**: global cells use $\gamma/\delta\gamma/\eta/\mu_{\mathrm{nc}}$ fixed at incremental-recursive posterior means (memo §5.1 convention).
3. **F2-full globals**: every parameter sampled freely in every cell. Fairest comparison.

2×2: paper-reported

	recursive	static	Δ(rec$-$sta)
incremental	-7180	-7199	$+19$ ($\Delta/\mathrm{dSE} = 10.28$)
global	-7628	-7626	-2 (null)
Δ(inc$-$glb)	$+448$	$+427$	
Δ/dSE	11.02	11.08	

Paper narrative:

- Architecture dominates: inc $\gg$ glb, $\Delta/\mathrm{dSE} \approx 11$.
- Recursive semantics helps incremental a lot, does *not* help global.
- Diff-in-diff $= +21$ (inc benefits more from rec).

2×2: F2 with fixed global parameters

	recursive	static	Δ(rec−sta)
incremental	-5294	-5297	$+2.80$ ($\Delta/\mathrm{dSE} = 2.22$, marginal)
global (fixed)	-5600	-5619	$+19.15$ ($\Delta/\mathrm{dSE} = 6.99$)
Δ(inc−glb)	$+306$	$+323$	
Δ/dSE	10.51	11.03	

- Architecture still dominant ($\Delta/\mathrm{dSE} \approx 10.5$).
- Interaction has flipped: rec now helps global, not incremental.
- But global's length parameters are *fixed at inc values* — not a fair test.

2×2: F2 with fully-sampled globals (fair)

	recursive	static	Δ(rec$-$sta)
incremental	-5294	-5297	$+2.80$ ($\Delta/\mathrm{dSE} = 2.22$)
global (full)	-5357	-5365	$+8.25$ ($\Delta/\mathrm{dSE} = 3.82$)
Δ(inc$-$glb)	$+63$	$+69$	
Δ/dSE	3.18	3.48	

Key changes vs paper-reported:

- Architecture effect Δ/dSE: $11 \to 3.2$ ($\sim 1/3$ of paper's).
- Regime-within-inc Δ/dSE: $10.28 \to 2.22$ ($\sim 1/5$ of paper's).
- Regime-within-glb: $0 \to 3.82$ (**appears, modestly**).
- Diff-in-diff: $+21 \to +5.45$ (same direction flipped, small magnitude).

Side-by-side

	Paper	F2-fixed	**F2-full**
R^2 (all cells)	.35	.94	.94
L1 residual	3.17	1.22	1.22
Lapse ε	0.22	0.10	0.10
Arch effect Δ/dSE	± 11	± 10.5	± 3.2
Regime $\|$inc Δ/dSE	10.28	2.22	2.22
Regime $\|$glb Δ/dSE	≈ 0	6.99	3.82
Interaction (diff-in-diff)	$+21$	-16	$+5.45$
Rec helps...	incremental	global (artefact)	slightly global

The tradeoff

Theory track (paper)

- Minimal, motivated mechanisms
- Clear 2×2 interaction
- Poor group-level fit ($R^2 = .35$)
- Seven large misfits in §5.3
- Lapse absorbs unexplained structure

Mechanism track (F2)

- Rich, but each term literature-grounded
- Excellent fit ($R^2 = .94$)
- All seven misfits resolved/improved
- Lapse halved ($0.22 \to 0.10$)
- Architecture signal shrinks to $\Delta/\text{dSE} = 3$
- Regime-within-inc weakens to marginal

Core tension. Mechanisms designed to reduce residuals explain variance that the paper's minimal parameterisation attributed to *architecture* $\times$ *regime*. The interaction survives in reduced form.

Two readings

Reading I: mechanisms as rivals

The paper's interaction was partly *architectural sparsity fitting error structure*. Once the error is absorbed by explicit mechanisms (colour salience, canonical ordering, context-dependent length), the residual architectural signal is modest. The main claim must weaken.

Reading II: mechanisms as scaffolding

λ_C, μ_{nc}, $\delta\gamma$, η are *implementations* of what context-updating $\times$ incremental architecture is *supposed to do*. Of course adding them absorbs architectural signal — they are the phenomena the architecture was indirectly capturing.

Both are defensible. The choice shapes the paper's framing.

Four concrete decisions

1. **Report v5 (F2) or keep the paper-reported model?**
   A middle path: main-text keeps the reported model; supplementary section shows F2 as mechanism-level refinement.
2. **If we adopt F2, how do we frame the interaction?**
   Reading I softens. Reading II preserves. A referee will ask.
3. **Adopt the corrected `COLOUR_SUFFICIENT` flag** ($ercf$, $zrdc$)**?**
   Independent of the F2 decision. Probably yes — the definition is simply more accurate.
4. **Generalise beyond the dc subset?**
   Parallel runs on df and cf subsets would test whether F2 mechanisms are symmetric or colour-specific.

Open questions

- **Strategy heterogeneity.** $\varepsilon = 0.10$ is smaller than ext-v1's 0.22 but non-zero. The `first/sharp` bare-`D` vs `DF` split looks like two participant sub-populations (minimum-description vs redundant-form). A mixture model is the natural next step.
- **Full-data ($N = 9586$) run.** Our early full-data run used the incorrect flag. A clean full-data F2 run would take $\sim$15 min on the GPU and complete the picture.
- **df and cf subsets.** Do the same mechanisms generalise? If a "form-sufficient" flag is needed, that would point to symmetric structure; if not, to colour-specificity.
- **LOO warnings.** All F2 `az.compare` outputs flagged `warning=True` (some Pareto $k > 0.7$). Differences reliable in direction, not in exact magnitude.

Summary

- Paper-reported model: theory-driven, interpretable 2×2 story, but $R^2 = .35$.
- Systematic residuals trace to three independent phenomena: **colour salience**, **canonical ordering**, **condition-dependent length bias**.
- Adding all three (v5 F2) lifts fit to $R^2 = .94$ and lapse to .10 with 11 free population parameters, every one informative, 0 divergences.
- Re-running the 2×2 under F2 with all parameters sampled: architecture effect shrinks $\sim 3\times$, regime effect shrinks and partially flips direction.
- Two readings of the tradeoff; both defensible.
- Decision the paper needs: **report F2, report paper-reported, or both?**

Artefacts: branch `feat/extension-v5`, spec at `docs/superpowers/specs/...`, plan at `docs/superpowers/plans/...`, synthesis at `10-writing/extension_v5_synthesis.md`.

Thanks

Questions and discussion.

`feat/extension-v5`	all changes, 15 commits
`10-writing/extension_v5_synthesis.md`	detailed write-up
`05-modelling-production-data/figures/v5/`	figures used above
`05-modelling-production-data/results/`	LOO + residual CSVs